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Special Article

Asian Working Group for Sarcopenia: 2019 Consensus Update on Sarcopenia Diagnosis and Treatment

Liang-Kung Chen MD, PhD [a,b,*], Jean Woo MD [c,**], Prasert Assantachai MD, PhD [d], Tung-Wai Auyeung MD [e], Ming-Yueh Chou MD [a,f], Katsuya Iijima MD, PhD [g], Hak Chul Jang MD, PhD [h], Lin Kang MD [i], Miji Kim PhD [j], Sunyoung Kim MD, PhD [k], Taro Kojima MD, PhD [l], Masafumi Kuzuya MD, PhD [m], Jenny S.W. Lee MD [e], Sang Yoon Lee MD, PhD [n,o], Wei-Ju Lee MD, MSc, PhD [a,p], Yunhwan Lee MD, MPH [q], Chih-Kuang Liang MD [a,f], Jae-Young Lim MD, PhD [n], Wee Shiong Lim MD [r], Li-Ning Peng MD, MSc, PhD [a,b], Ken Sugimoto MD, PhD [s], Tomoki Tanaka PhD [f], Chang Won Won MD, PhD [k], Minoru Yamada PhD [t], Teimei Zhang PhD [u], Masahiro Akishita MD, PhD [l], Hidenori Arai MD, PhD [v,***]

[a] *Aging and Health Research Center, National Yang Ming University, Taipei, Taiwan*
[b] *Center for Geriatrics and Gerontology, Taipei Veterans General Hospital, Taipei, Taiwan*
[c] *Department of Medicine and Therapeutics, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong S.A.R., China*
[d] *Division of Geriatric Medicine, Department of Preventive and Social Medicine, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand*
[e] *The S. H. Ho Center for Gerontology and Geriatrics, The Chinese University of Hong Kong, Hong Kong S.A.R, China*
[f] *Center for Geriatrics and Gerontology, Kaohsiung Veterans General Hospital, Kaohsiung, Taiwan*
[g] *Institute of Gerontology, The University of Tokyo, Tokyo, Japan*
[h] *Department of Internal Medicine, Seoul National University College of Medicine, Seoul National University Bundang Hospital, Seongnam, Korea*
[i] *Department of Geriatrics, Peking Union Medical College Hospital, Beijing, China*
[j] *Department of Biomedical Science and Technology, College of Medicine, East-West Medical Research Institute, Kyung Hee University, Seoul, Korea*
[k] *Department of Family Medicine, College of Medicine, Kyung Hee University, Seoul, Korea*
[l] *Department of Geriatric Medicine, Graduate School of Medicine, The University of Tokyo, Tokyo, Japan*
[m] *Department of Community Healthcare and Geriatrics, Nagoya University Graduate School of Medicine, Aichi, Japan*
[n] *Department of Rehabilitation Medicine, Seoul National University College of Medicine, Seoul National University Bundang Hospital, Seongnam, Korea*
[o] *Department of Rehabilitation Medicine, Seoul National University Boramae Medical Center, Seoul, Korea*
[p] *Department of Family Medicine, Taipei Veterans General Hospital Yuanshan Branch, Yilan, Taiwan*
[q] *Department of Preventive Medicine and Public Health, Ajou University School of Medicine, Suwon, Korea*
[r] *Institute of Geriatrics and Active Aging, Tan Tock Seng Hospital, Singapore*
[s] *Department of Geriatric Medicine, Osaka University Graduate School of Medicine, Osaka, Japan*
[t] *Graduate School of Comprehensive Human Sciences, University of Tsukuba, Tokyo, Japan*
[u] *Beijing Institute of Geriatrics, Beijing Hospital, Ministry of Health, Beijing, China*
[v] *National Center for Geriatrics and Gerontology, Obu, Aichi, Japan*

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* Address correspondence to Liang-Kung Chen, MD, PhD, Center for Geriatrics and Gerontology, Taipei Veterans General Hospital, No. 201, Sec 2, Shih-Pai Road, Taipei 11217, Taiwan.

** Jean Woo, MD, Department of Medicine and Therapeutics, 9F Lui Che Woo Clinical Science Administration Building, Prince of Wales Hospital, Ngan Shing St., Shatin, NT, Hong Kong S.A.R, China.

*** Hidenori Arai, MD, PhD, National Center for Geriatrics and Gerontology, 7-430 Morioka-cho, Obu, Aichi 474-0038, Japan.

E-mail addresses: lkchen2@vghtpe.gov.tw (L.-K. Chen), jeanwoowong@cuhk.edu.hk (J. Woo), harai@ncgg.go.jp (H. Arai).

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A B S T R A C T

Clinical and research interest in sarcopenia has burgeoned internationally, Asia included. The Asian Working Group for Sarcopenia (AWGS) 2014 consensus defined *sarcopenia* as "age-related loss of muscle mass, plus low muscle strength, and/or low physical performance" and specified cutoffs for each diagnostic component; research in Asia consequently flourished, prompting this update. AWGS 2019 retains the previous definition of sarcopenia but revises the diagnostic algorithm, protocols, and some criteria: low muscle strength is defined as handgrip strength <28 kg for men and <18 kg for women; criteria for low physical performance are 6-m walk <1.0 m/s, Short Physical Performance Battery score ≤ 9, or 5-time chair stand test ≥ 12 seconds. AWGS 2019 retains the original cutoffs for height-adjusted muscle mass: dual-energy X-ray absorptiometry, <7.0 kg/m^2 in men and <5.4 kg/m^2 in women; and bioimpedance, <7.0 kg/m^2 in men and <5.7 kg/m^2 in women. In addition, the AWGS 2019 update proposes separate algorithms for community vs hospital settings, which both begin by screening either calf circumference (<34 cm in men, <33 cm in women), SARC-F (≥ 4), or SARC-CalF (≥ 11), to facilitate earlier identification of people at risk for sarcopenia. Although skeletal muscle strength and mass are both still considered fundamental to a definitive clinical diagnosis, AWGS 2019 also introduces "possible sarcopenia," defined by either low muscle strength or low physical performance only, specifically for use in primary health care or community-based health promotion, to enable earlier lifestyle interventions. Although defining sarcopenia by body mass index–adjusted muscle mass instead of height-adjusted muscle mass may predict adverse outcomes better, more evidence is needed before changing current recommendations. Lifestyle interventions, especially exercise and nutritional supplementation, prevail as mainstays of treatment. Further research is needed to investigate potential long-term benefits of lifestyle interventions, nutritional supplements, or pharmacotherapy for sarcopenia in Asians.

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Age-related loss of skeletal muscle, termed "sarcopenia," has grave physiological and clinical consequences.[1–4] The original operational definition compared loss of appendicular skeletal muscle mass (ASM) in older adults relative to sex-specific ASM measurements of 20- to 40-year-olds; however, there was no universal definition of muscle strength.[1,2] In 2010, the European Working Group on Sarcopenia in Older People (EWGSOP) proposed a diagnostic algorithm for sarcopenia, in which both muscle quantity and quality were cardinal requirements.[4] Subsequently, sarcopenia research burgeoned worldwide,[2] Asia being no exception. However, diagnosing sarcopenia in Asian people requires some special considerations because of anthropometric and cultural or lifestyle-related differences compared with Western contemporaries; for example, relatively smaller body size, higher adiposity, and less mechanized, more physically active lifetimes.[5]

In 2014, the Asian Working Group for Sarcopenia (AWGS) proposed a diagnostic algorithm based on Asian data; this resembled EWGSOP 2010 but, besides measurement protocols, clearly defined cutoffs for individual diagnostic components.[5] AWGS 2014 invigorated sarcopenia research in Asia, which augmented the evidence base, and ongoing AWGS discussions consolidated the challenges and questions in sarcopenia. The AWGS[6] reported important advances in 2016 and planned to update its consensus accordingly.

Sarcopenia was assigned an individual International Statistical Classification of Diseases and Related Health Problems code (M62.84) in 2016, which stimulated diagnostic and therapeutic trials internationally. Japan published clinical practice guidelines for sarcopenia in 2018,[7] which systematically reviewed the latest evidence and promoted awareness. The EWGSOP issued an updated consensus in 2019 (EWGSOP2).[8]

Consensus Process

AWGS founding members (L. K. Chen, H. Arai, J. Woo, C. W. Won, and P. Assantachai) used PubMed searches to identify other researchers with pertinent expertise from Asian countries or regions, whom they invited to lead special interest groups on the topics of epidemiology, case-finding, diagnosis, muscle mass, muscle strength, physical performance, and treatment. Group leaders were appointed at face-to-face meetings in Nagoya, Japan, and tasked to recruit members to help them source and appraise evidence in their respective topic areas. Each special interest group then reviewed research from East and Southeast Asia* published since AWGS 2014. Twenty-five AWGS members participated in a consensus meeting held in Hong, during May 2019, where each special interest group reported their findings and the updated AWGS consensus was decided unanimously. Evidence on sarcopenia interventions was largely based on the Japanese guideline,[9] which used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology, supplemented by more recent publications.

Definition of Sarcopenia

The definition and diagnosis of sarcopenia are still evolving as new findings challenge current understanding. Nevertheless, given limited Asian data and controversy about using the term *sarcopenia* where muscle wasting may predominantly be due to coexisting conditions such as cachexia or paralysis, the AWGS decided against following a recent trend to consider all muscle wasting as sarcopenia and therefore retains its original definition of "age-related loss of skeletal muscle mass plus loss of muscle strength and/or reduced physical performance," without reference to comorbidity. AWGS 2019 retains the age cutoffs at either 60 or 65 years old, depending on how each country defines "older people"; although sarcopenic features may occur in younger adults, the underlying pathophysiology should be investigated rather than simply pursuing a diagnosis of sarcopenia. The AWGS supports early identification of people at risk, to enable timely intervention, but does not endorse the concept of "pre-sarcopenia," because of insufficient evidence of its prognostic value. The AWGS recommends necessary interventions for sarcopenia with or without specific contributing clinical conditions. For people with uncontrolled or acute clinical conditions that may lead to sarcopenia, the

* This AWGS 2019 consensus is based on expert knowledge and research evidence from East and Southeast Asian countries/regions including China, Hong Kong, Japan, Singapore, South Korea, Taiwan, and Thailand.

AWGS recommends proper interventions to prevent sarcopenia from developing or deteriorating, concurrent with treating underlying clinical conditions. Given that age-related changes in different aspects of body composition may have different clinicopathologic consequences in older adults, comorbid conditions such as sarcopenic obesity, osteosarcopenia, sarco-osteoporosis, or osteosarcopenic obesity may exert synergistic effects.[3,10,11] Although the AWGS supports research efforts to investigate this "double burden" hypothesis, we will not recommend specific diagnostic criteria or treatments unless and until more convincing evidence becomes available.

Epidemiology

The review of epidemiology studies from Asian countries that used AWGS 2014 criteria discovered that the prevalence of sarcopenia ranged from 5.5% to 25.7%, with male predominance (5.1%-21.0% in men vs 4.1%-16.3% in women) (Supplementary Table 1)[12–14,S1–S10]; the prevalence became more consistent—7.3% to 12.0%—if only studies with more than 1000 male and female participants were included.[12–14,S9] Older age may be the most important among numerous reported risk factors, but household status, lifestyle, physical inactivity, poor nutritional and dental status, and diseases (osteoporosis, metabolic diseases, etc) were also independently associated with sarcopenia (Supplementary Table 2). In particular, the likelihood of developing sarcopenia is significantly correlated with the number of cardiometabolic risk factors, notably diabetes, hypertension, and dyslipidemia. The pathogenesis of sarcopenia may involve satellite-cell senescence, loss of motor neurons, less active neuromuscular junctions, hormonal status, pro-inflammatory cytokines, decreased mitochondrial function, abnormal myokine production, and weight loss accompanying decreased appetite.[2–4]

In terms of longer-term clinical outcomes, AWGS-defined sarcopenia was significantly associated with increased risks of physical limitations at 4 years, slowness at 7 years, and 10-year mortality, but not of hospitalization.[13–15] In 2 observational studies of older Chinese people, sarcopenia substantially increased the risk of fractures, and improved the efficacy of FRAX in predicting incident fractures in men, but not in women.[12–15] In a 4-year follow-up study of 1099 older Japanese people, osteoporosis was associated with the development of sarcopenia, but no reciprocal relationship was evident.[12]

Diagnosis of Sarcopenia

Figure 1 shows the AWGS 2019 algorithm for identifying and diagnosing older adults with or at-risk for sarcopenia, including case-finding, assessment, and diagnostic protocols for use in either hospital and research settings, or in community-based health care and screening.

The special interest group review of sarcopenia diagnosis discovered that most studies done in Asian countries had used the AWGS 2014 criteria. Following extensive deliberations, AWGS 2019 contends that diagnosing sarcopenia requires measurements of both muscle quality and quantity and defines persons with low muscle mass, low muscle strength, and low physical performance as having "severe sarcopenia."

Acknowledging the difficulty of measuring muscle mass in community settings, AWGS 2014 supported ASM measurement using bioelectrical impedance analysis (BIA). AWGS 2019 recommends

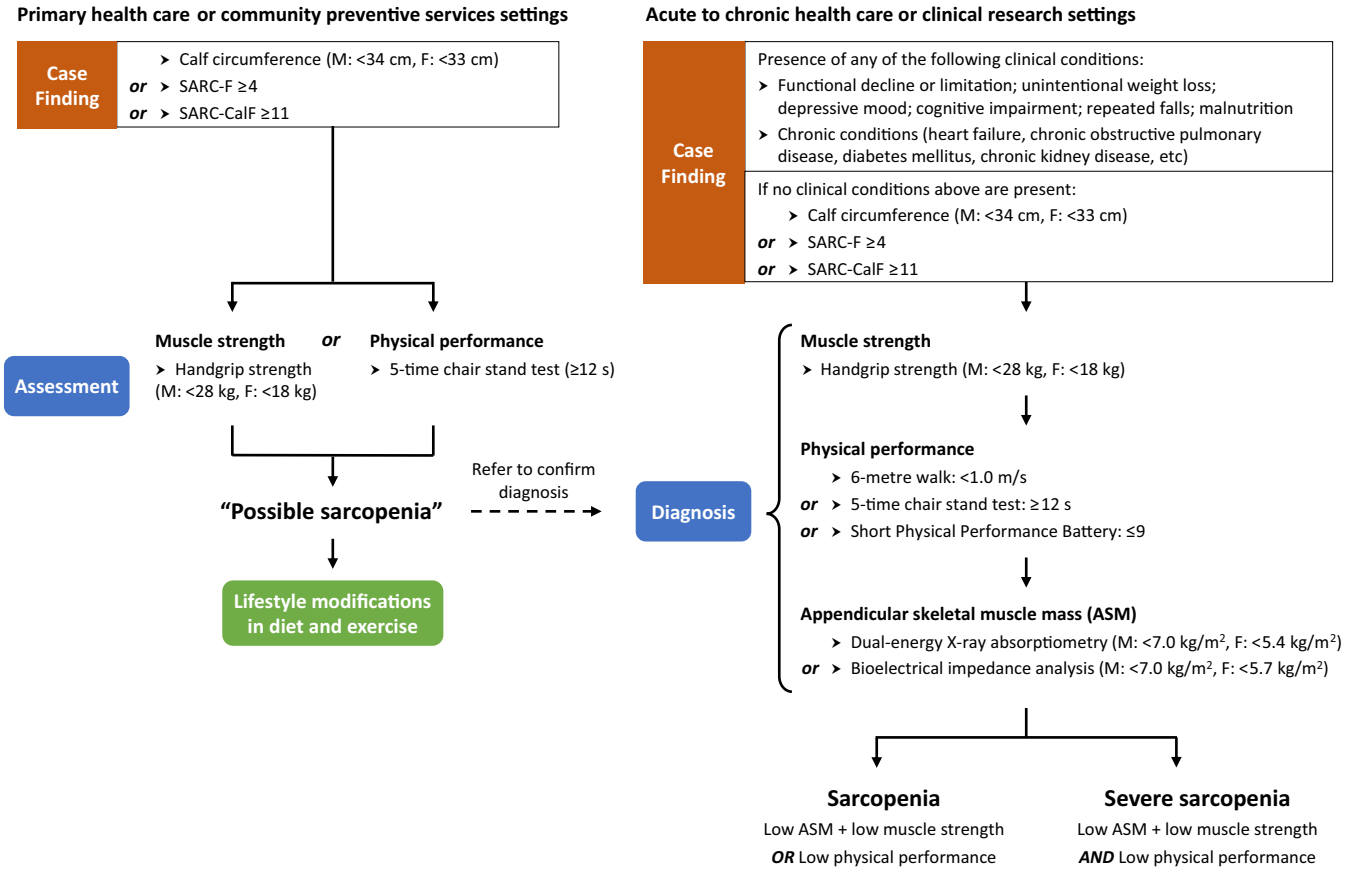


Fig. 1. AWGS 2019 algorithm for sarcopenia. F, female; M, male.

further strategies for early identification of people with, or at risk for, sarcopenia, to facilitate necessary interventions in settings without advanced diagnostic equipment. Specifically, AWGS 2019 introduces "possible sarcopenia," defined by low muscle strength with or without reduced physical performance, which is recommended for use in primary health care and preventive services, but not in hospital or research settings.

The AWGS recommends lifestyle interventions and related health education for primary health care and preventive service users with "possible sarcopenia" but also encourages their referral to hospital for confirmatory diagnosis; lifestyle interventions should continue regardless of the final diagnosis. In hospital and research settings, besides following the diagnostic protocols, health care professionals should investigate potential underlying causes, especially reversible ones, and provide appropriate personalized intervention programs for older adults with sarcopenia.

Case-finding

The case-finding special interest group reviewed studies from East and Southeast Asia that used screening or assessment instruments to identify early signs of sarcopenia. Based on related validation reports, AWGS 2019 recommends using either calf circumference (CC) or the SARC-F or SARC-CalF questionnaires for case-finding.

The recommended protocol for CC measures the maximum value of both calves using a nonelastic tape, which has moderate-to-high sensitivity and specificity in predicting sarcopenia or low skeletal muscle mass.[16–20] Reported CC cutoffs were 32 to 34 cm in men and 32 to 33 cm in women; AWGS 2019 recommends CC $<$34 cm for men and $<$33 cm for women for screening or case-finding. An effective alternative to CC is the "Yubi-wakka" (finger-ring) test, in which people use the index fingers and thumbs of both hands to encircle the thickest part of their nondominant calf; older adults are at increased risk for sarcopenia if the measured calf just fits or is smaller than their finger-ring.[21]

The SARC-F questionnaire assesses 5 components: strength, assistance in walking, rising from a chair, climbing stairs, and falls.[22] Studies in Asia have validated different language versions of SARC-F[23–27] and have shown that the results are independently associated with adverse clinical outcomes.[28–33] A SARC-F score $\geq$4 was defined as sarcopenia, and the low sensitivity and high specificity of SARC-F may facilitate the referral of at-risk persons for definitive diagnosis rather than community-based follow-up.[4,23–27] SARC-CalF improved the sensitivity of SARC-F by adding CC, with a score $\geq$11 diagnosing sarcopenia.[34]

Skeletal Muscle Mass Measurement

The special interest group review on muscle mass measurement focused on techniques, body-size adjustment methods, and determining cutoff values. Although magnetic resonance imaging, computed tomography, dual-energy X-ray absorptiometry (DXA), and BIA have all been used to estimate skeletal muscle mass, DXA and BIA were the most commonly used modalities in Asia, and the AWGS 2014 cutoffs were widely adopted. BIA measurement with a multifrequency device correlated more closely with DXA-measured ASM than did BIA measured with other devices.[35] AWGS 2019 does not recommend BIA devices designed for home use because of suboptimal diagnostic accuracy.[36] Although some studies report ultrasonographic assessment of lower leg muscle to estimate muscle mass,[37,38] the AWGS does not recommend ultrasonography until more data become available. Hence, AWGS 2019 recommends using either DXA or multifrequency BIA, both height-adjusted, for measuring muscle mass in sarcopenia diagnosis.

Table 1 summarizes muscle mass measurement protocols and related outcome indicators in studies from East and Southeast Asia.[39,S1,S3,S11–S35] Previous research indicates that body mass index (BMI)–adjusted muscle mass may be superior to unadjusted muscle mass in predicting functional outcomes and disability in older adults.[40] In a longitudinal study, sarcopenia defined according to US Foundation for the National Institutes of Health criteria was more strongly associated with adverse clinical outcomes than was height-adjusted skeletal muscle mass[41]; however, more studies are needed before formally endorsing the BMI-adjusted US Foundation for the National Institutes of Health diagnostic cutoffs. Consequently, AWGS 2019 encourages studies using BMI-adjusted muscle mass to determine the best way to measure muscle mass.

The AWGS 2019 cutoffs for low muscle mass in sarcopenia diagnosis are as follows: $<$7.0 kg/m^2 in men and $<$5.4 kg/m^2 in women by DXA; and $<$7.0 kg/m^2 in men and $<$5.7 kg/m^2 in women by BIA. If US Foundation for the National Institutes of Health criteria are used, $<$0.789 kg/BMI for men and $<$0.512 kg/BMI for women may be appropriate cutoffs (DXA-measured muscle mass only).

Muscle Strength

Almost all studies in East and Southeast Asia that focused on muscle strength, function, and sarcopenia had used handgrip strength to indicate skeletal muscle strength; therefore, AWGS 2019 retains the recommendation to use handgrip strength. Although handgrip strength was widely used, measurement protocols varied between studies. The devices used most often in Asia are the spring-type dynamometer (Smedley), followed by hydraulic-type (Jamar);[42] AWGS 2019 recommends using either device to diagnose sarcopenia, provided the standard protocols for the specific model are followed. Pertinently, data generated from these dynamometers are not directly comparable, because the handgrip strength of an older adult measured using the Jamar hydraulic dynamometer may be higher than that indicated by the Smedley dynamometer. AWGS 2019 does not propose dynamometer-specific cutoff values because of insufficient comparative data.[42]

The American Society of Hand Therapists and the Southampton protocol recommend sitting with 90° elbow flexion as the standard position for measuring handgrip strength using the Jamar dynamometer;[43] however, the US National Health and Nutrition Examination Survey (NHANES) protocol for using the Smedley dynamometer recommends standing with full elbow extension, similar to the Canadian Health Measures Survey protocol.[44] The NHANES protocol also permits sitting for people who are unable to stand unassisted. In selected review articles with clear descriptions about positioning, standing with full elbow extension was the most common approach when using the Smedley dynamometer, whereas most measurements with the Jamar dynamometer were taken from seated subjects with 90° elbow flexion. Therefore, the standard positionings recommended by AWGS 2019 are standing with full elbow extension for the Smedley dynamometer and sitting with 90° elbow flexion for the Jamar dynamometer. Nevertheless, sitting is preferable if older people cannot stand unassisted. Most published studies measured both hands or the dominant hand for muscle strength and used the best performance of either 2 or 3 trials. The handgrip strength measurement protocol recommended by AWGS 2019 is to take the maximum reading of at least 2 trials using either both hands or the dominant hand in a maximum-effort isometric contraction, rather than using a fixed acquisition time.

Most published Asian studies defined low handgrip strength according to the AWGS 2014 cutoffs, which were primarily based on 2 studies in Asia. More published and unpublished data from community-based cohorts in East and Southeast Asia have since become available, and the AWGS has revised some cutoffs for

Table 1
Measurement of Muscle Mass in Sarcopenia Diagnosis and Its Outcomes and Associated Factors

Design (Participants)	Sarcopenia Criteria	ASM Measurement		Cutoff Values		Associated Factors and Outcomes	Data Source: Author, Year[ref]
		Technique	Adjustment	Men	Women		
Cross-sectional (community-dwellers)	AWGS	BIA: InBody S10, 370, 430, 720, 770	ASM/Ht^2	<7.0	<5.7	Arterial stiffness	Zhang, 2019[S11]
						Low circulating irisin level	Chang, 2017[S12]
						Household status and locomotive syndrome	Momoki, 2017[S1]
						Depressive symptoms	Wang, 2018[S3]
						High plasma proline concentration	Toyoshima, 2017[S13]
		DXA: Discovery-W; QDR 4500R, 4500A; Lunar iDXA; iDXA GE		<7.0	<5.4	Frailty	Choe, 2017[S14]
						Hip bone strength indices	Choi, 2018[S15]
						Cardiac function	Sugie, 2017[S16]
						Advanced glycation end-product accumulation	Kato, 2017[S17]
	FNIH		ASM/BMI	<0.789	<0.512	Frailty	Choe, 2017[S14]
						Albuminuria	Han, 2016[S18]
	ASM/Wt (kg) ≥1 SD below YAM		ASM/Wt (%)	<30.8	<24.3	Metabolic phenotype: metabolic syndrome, hypertension, diabetes, or cardiovascular disease	Hwang, 2017[S19]
				<30.1	<21.2	Multimorbidity	An, 2016[S20]
	ASM/Ht^2 >2 SD below YAM		ASM/Ht^2	<6.52		Higher white blood cell count and lower vitamin D levels	Kim, 2017[S21]
Cross-sectional (consecutive inpatients)	AWGS	BIA: InBody S10, 720, 770; Tanita MC-980A	ASM/Ht^2	<7.0	<5.7	Blood pressure variability	Hashimoto, 2018[S22]
						Poor oral status	Shiraishi, 2018[S23]
						Ambulatory status	Maeda, 2017[S24]
						Back muscle strength	Toyoda, 2019[S25]
						Lower bone mineral content, BMI, serum albumin	Fukuoka, 2019[S26]
		DXA: Discovery-W		<7.0	<5.4	Malnutrition	Tay, 2018[S27]
						Pre-albumin levels	Chen, 2019[S28]
						Non-alcoholic fatty liver disease	Zhai, 2018[S29]
Retrospective (consecutive inpatients)	AWGS	BIA: InBody S10	ASM/Ht^2	<7.0	<5.7	Worse recovery of activities of daily living, dysphagia and lower home discharge	Yoshimura, 2019[S30]
		DXA: DPX-NT		<7.0	<5.4	Malnutrition, chronic inflammation and 1-y mortality	Yoo, 2018[S31]
Prospective cohort (community-dwellers)	AWGS	BIA: InBody	ASM/Ht^2	<7.0	<5.7	Depressive symptoms	Chen, 2019[S32]
		DXA: ODR-4500		<7.0	<5.4	All-cause mortality	Yuki, 2017[39]
Prospective cohort (inpatients)	AWGS	BIA: InBody S10	ASM/Ht^2	<7.0	<5.7	90-day mortality	Maeda, 2017[S33]
	Lowest quintile	Formula using Wt, Ht, sex, and age*		<6.92	<5.13	3-year mortality	Hu, 2017[S34]
				<6.70	<4.75	Hospital readmission and 3-y mortality	Yang, 2017[S35]

FNIH, Foundation for the National Institutes of Health; Ht, height (m); SD, standard deviation; Wt, body weight (kg); YAM, young adult mean.
*$ASM = 0.193 \times bodyweight + 0.107 \times height - 4.157 \times gender - 0.037 \times age - 2.631$.

Table 2
Measurement of Physical Performance for Sarcopenia Diagnosis and Gait Speed Measurement Protocols

Study Design	Types of Physical Performance	Gait Speed Measurement Protocols				Data Sources: Author, Year[ref]
		Starting Method	Usual or Fastest Speed	Walk Distance, m	Cutoff, m/s	
Cross-sectional	Gait speed only	Moving	Usual	6.0	0.8	Tanishima, 2017[S5]; Toyoshima, 2017[S13]; Fukuoka, 2019[S26]
				5.0	0.8	Matsumoto, 2019[46]; Ida, 2017[S36]
				5.0	1.26	Ishii, 2016[47]
			NR	5.0	0.8	Toyoda, 2019[S25]
		NR	Usual	6.0	0.8	Wang, 2018[S3]; Chen, 2019[S28]; Zhai, 2018[S29]; Lu, 2019[S37]
				5.0	NR	Yang, 2017[S38]
				4.0	0.8	Meng, 2017[S39]
				10.0	0.8	Yanishi, 2017[S40]
			NR	6.0	0.8	Yang, 2018[24]; Yang, 2019[25]; Yang, 2018[S41]; Zeng, 2018[S42]
				10.0	0.8	Yanishi, 2017[S43]
				4.5	NR	Abdul Aziz, 2018[S44]
	Gait speed and SPPB	NR	Usual	4.0	0.8	Tanaka, 2017[33]; Kamo, 2018[S45]
			NR	3.0	0.8	Tay, 2018[S27]
	Gait speed, TUG, and one-leg standing	Moving	Fastest	5.0	1.0	Kera, 2017[48]
Retrospective	Gait speed only	Moving	Usual	10.0	0.8	Yuki, 2017[39]
		Static	Usual	6.0	0.8	Lou, 2017[S46]
		NR	Usual	6.0	0.8	Roh, 2017[S47]
	Gait speed and SPPB	Moving	Usual	4.0	NR	Jung 2018[S48]
	SPPB					Makiura, 2016[49]; Kim, 2018[S49]
Prospective cohort	Gait speed only	Moving	Usual	4.0	0.8	Jang, 2018[S50]
		NR	Usual	4.0	0.8	Yang, 2017[S35]; Moon, 2016[S51]
				6.0	0.8	Roh, 2017[S52]
			NR	NR	0.8	Harimoto, 2017[S53]
	Gait speed and SPPB	Moving	Usual	4.0	0.8	Kim, 2018[23]
Randomized controlled trial	Gait speed only	Moving	Usual	5.0	NR	Kim, 2016[S54]
		NR	NR	NR	0.8	Mori, 2018[50]
	Gait speed and SPPB	NR	Usual	6.0	0.8	Shen, 2016[S55]
	SPPB					Jang, 2018[S56]
	Senior Fitness Test					Hong, 2017[S57]

NR, not reported; TUG, timed-up-and-go.

handgrip strength accordingly; these remain based on the lowest quintile, as this approach may also facilitate frailty research in Asia. Having analyzed data from 8 Asian cohorts comprising 21,984 participants aged ≥65 years, the AWGS 2019 recommends low muscle strength diagnostic cutoffs of handgrip <28.0 kg for men and <18.0 kg for women.[45]

Physical Performance

Published research used a wide range of physical performance tests, including the Short Physical Performance Battery (SPPB), usual gait speed, 6-minute walk test, stair-climb power test, timed-up-and-go test, and 5-time chair stand test.[4–6] Usual gait speed was the most frequently used test and was strongly associated with the onset of disability, severe mobility limitation, and mortality (Table 2).[23–25,33,46–48,S3,S5,S13,S25–S29,S35–S57] AWGS 2019 recommends defining low physical performance based on either SPPB, 6-m walk, or 5-time chair stand test cutoffs; timed-up-and-go is not recommended, because the results may reflect multiple complex pathoetiologies. Protocols for measuring gait speed in published Asian studies had diverse methodologies (Table 2); AWGS 2019 recommends measuring the time taken to walk 6 m at a normal pace from a moving start, without deceleration, and taking the average result of at least 2 trials as the recorded speed. Although most studies used a manual stopwatch, automatic timing is increasingly used.[39,46]

Most Asian studies used the AWGS 2014 cutoff of ≤0.8 m/s for sarcopenia diagnosis, but some used other criteria based on their own cohorts or previous publications.[47,48,51] Slow gait speed defined as <1 m/s independently predicted rapid cognitive decline in oldest-old men and higher dementia risk among community-dwelling older adults.[52] In fact, many frailty or functional disability studies define slowness as 6-m walk <1.0 m/s.[2,51] In a recent study of older adults with sarcopenia based on the AWGS 2014 criteria, the mean gait speed was 0.96 m/s.[53] Other Asian data (including gray literature) showed a similar lowest quintile for gait speed in study cohorts from several different countries. Therefore, AWGS 2019 recommends using usual gait speed to define reduced physical performance and has increased the cutoff from ≤0.8 to <1.0 m/s. Although some studies have reported sex differences in gait speed, AWGS 2019 does not recommend sex-specific cutoffs.

EWGSOP2 defined low physical performance as total SPPB score ≤8.[8] However, impaired mobility defined as SPPB score ≤9 was more predictive of all-cause mortality in a systematic review[54]; the mean SPPB scores of older adults with AWGS-defined sarcopenia in a recent report was 10.5.[49] Therefore, AWGS 2019 recommends SPPB score ≤9 as the cutoff for low physical performance. Additionally, 5-time chair stand time has been proposed as a surrogate for gait speed in sarcopenia diagnosis.[55] A 5-time chair stand time cutoff of 11.6 seconds corresponded to a walking speed of 1.0 m/s; hence, AWGS 2019 recommends ≥12 seconds as the cutoff for low physical performance.

Treatment and Intervention

A systematic review of randomized controlled trials (RCTs) through 2016 examined the effects of exercise, nutrition, combined exercise and nutrition, and pharmacologic agents on sarcopenia.[9] For people with sarcopenia as defined by EWGSOP 2010, AWGS 2014, or combined muscle mass and function and/or strength, exercise alone

effectively increased ASM and gait speed. Meta-analysis of 12 RCTs (not exclusively using AWGS 2014 or EWGSOP 2010 criteria) showed variable effects of nutrition alone on muscle mass, strength, and gait speed.[9] Interventions focused on the intake of essential amino acids may improve knee extension in patients with sarcopenia. Although combined interventions improved sarcopenia in RCTs, longer-term effects beyond 3 months were uncertain. There were no differences between combined interventions vs either exercise or nutrition only. Selective androgen receptor modulators are partially effective in increasing muscle mass but not strength or function.[9,56] Early trials suggest that myostatin neutralizing antibody or activin IIB receptor blockade may significantly increase ASM and aspects of functional performance, but the potential clinical benefits remain uncertain.[56]

A supplementary search of literature on interventions in Asian populations for years 2017-2018 identified 4 RCTs that had used the AWGS criteria for sarcopenia and evaluated both exercise and nutrition; 2 were hospital based[57,58] and 2 studied healthy older community residents.[50,59] Hospital-based interventions, which included resistance exercise plus nutritional supplements including branched-chain amino acids, vitamin D, whey protein, and hydroxymethylbutyrate (HMB) enriched milk, significantly increase physical function, muscle mass, and strength. In a 24-week study of community-living women in Japan, those taking whey protein after resistance exercise had increased muscle mass, grip strength, and gait speed compared with those in either single intervention arm.[50] Another study compared community-living men and women randomized to either 12 weeks of exercise, exercise plus HMB-enriched supplement, or a wait-listed control group; there was no effect on the primary outcome of gait speed, but both intervention groups had improved leg extension and 5-time chair stand performance, which persisted for 12 weeks after the intervention ended. Leg muscle mass and ASM only increased in the exercise plus nutritional supplementation group; however, the increase in muscle mass had disappeared by 24 weeks.[59]

Available evidence suggests that exercise plus nutrition improves muscle strength and function, with variable effects on muscle mass. However, there is uncertainty about the desired treatment outcome indicators for intervention: whether the goal should be changing sarcopenia status (reversal from sarcopenic to nonsarcopenic) or changing the individual components (muscle mass, strength, and physical performance) that define sarcopenia to above their respective cutoffs. Furthermore, the outcomes desired in community vs hospital settings may differ. Last but not least, it would be relevant to include patient-related outcome measures, such as general or specific quality-of-life measures[60,61] or even self-rated health. Ongoing pharmacotherapy trials are evaluating activin receptor antagonist, myostatin or activin inhibitor, androgen receptor modulators, and troponin activator of fast skeletal muscle.[56]

Implications for Practice, Policy, and Research

AWGS 2019 highlights the impact of sarcopenia in all health care settings and recommends individualized lifestyle intervention that may be implemented across the health care spectrum. To promote healthy aging, AWGS 2019 has defined a new entity of “possible sarcopenia” to facilitate timely lifestyle intervention in community health care and prevention settings, which will contribute to higher awareness of sarcopenia prevention and interventions in diverse health care settings. However, more intervention studies are needed to optimize lifestyle, nutritional, or pharmaceutical interventions for Asian people with sarcopenia.

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Appendix

Supplementary Table 1
Prevalence of Sarcopenia Defined by the Asian Working Group for Sarcopenia 2014 Criteria

Country/Region (Author, Year)[ref]	Muscle Mass Measurement	Age Cutoff, y	Sample Size	Prevalence, %		
				Total	Men	Women
Japan (Yoshimura, 2017)[12]	DXA	60	1099	8.2	8.5	8.0
Japan (Momoki, 2017)[S1]	BIA	65	186	NR	NR	21.0
Japan (Iwasaki, 2017)[S2]	BIA	75	272	25.7	NR	NR
China (Wang, 2018)[S3]	BIA	60	948	7.1	6.6	7.5
China (Hai, 2017)[S4]	BIA	60	834	10.6	11.3	9.8
China (Yu, 2014)[15]	DXA	65	4000	7.3	9.4	5.3
Japan (Tanishima, 2017)[S5]	BIA	40	216	5.5	6.3	5.1
China (Han, 2017)[S6]	BIA	60	711	10.8	8.3	13.3
Hong Kong (Woo 2015)[14]	DXA	65	2000	9.4	9.4	NR
China (Hai, 2017)[S7]	BIA	60	836	10.5	11.3	9.7
China (Hu, 2017)[S8]	DXA	60	607	18.5	16.3	19.9
China (Wang, 2016)[S9]	BIA	60	1090	12.0	14.0	10.2
Taiwan (Kuo, 2019)[S10]	DXA	60	731	6.8	9.3	4.1

DXA, dual X-ray absorptiometry; BIA, bioelectrical impedance analysis; NR, not reported.

Supplementary Table 2
Risk Factors of Sarcopenia in Studies From East and Southeast Asia Since 2014

Categories	Risk Factors
Demographic characteristics	Age, sex
Household status	Living alone or living with children and/or grandchildren; Person's satisfaction with their perceived level of family function (family APGAR score)
Lifestyle habits	Binge drinkers with weekly or daily alcohol consumption (women); short sleep duration or having long sleep duration (women); water intake from food (g/d and cup/d) and dietary water adequacy ratio (mL)
Physical activity	Locomotive syndrome (one study for women)
Dietary pattern, dental condition and nutritional status	Lower frequency of nut consumption per week; impaired dentition status; higher dietary variety score (one study for women); lower body mass index (<18.5); risk of malnutrition (MNA score)
Comorbidities	Osteoporosis; cardiovascular risk factors (including type 2 diabetes mellitus, hypertension, dyslipidemia)

APGAR, adaptability, partnership, growth, affection, and resolve; MNA, Mini Nutritional Assessment.

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